# 55. Capstone: Secure Edge for a Small App

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**Phase:** Phase 13 — Capstones
**Estimated time:** 4–6 days
**Slug:** `55-capstone-edge`

### Learning objectives

- Terminate TLS at edge
- Reverse proxy to app
- Firewall + SSH hardening + health checks
- Probe latency/error metrics

### Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

### Mission

Deploy a small app behind:

- Reverse proxy with TLS (lab certs OK)
- Health checks and structured logs
- Host firewall
- Optional WireGuard admin access instead of public SSH
- Simple uptime probe script with curl timings
- Runbook for cert renewal and failure modes

### Rubric

- Least privilege exposure
- Clear 502/504 debugging notes
- Evidence of probe dashboard/log
- Graceful reloads

### Common mistakes

- App exposed directly on :80 public
- No health checks
- Certs without renewal plan

### Exercises

Work in `exercises/55-capstone-edge/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

**Do not open `solutions/55-capstone-edge/` until after an honest attempt.**

## Self-check

1. What is public?
2. How detect expiry?
3. How prove health?

## Further reading

- https://nginx.org/en/docs/